

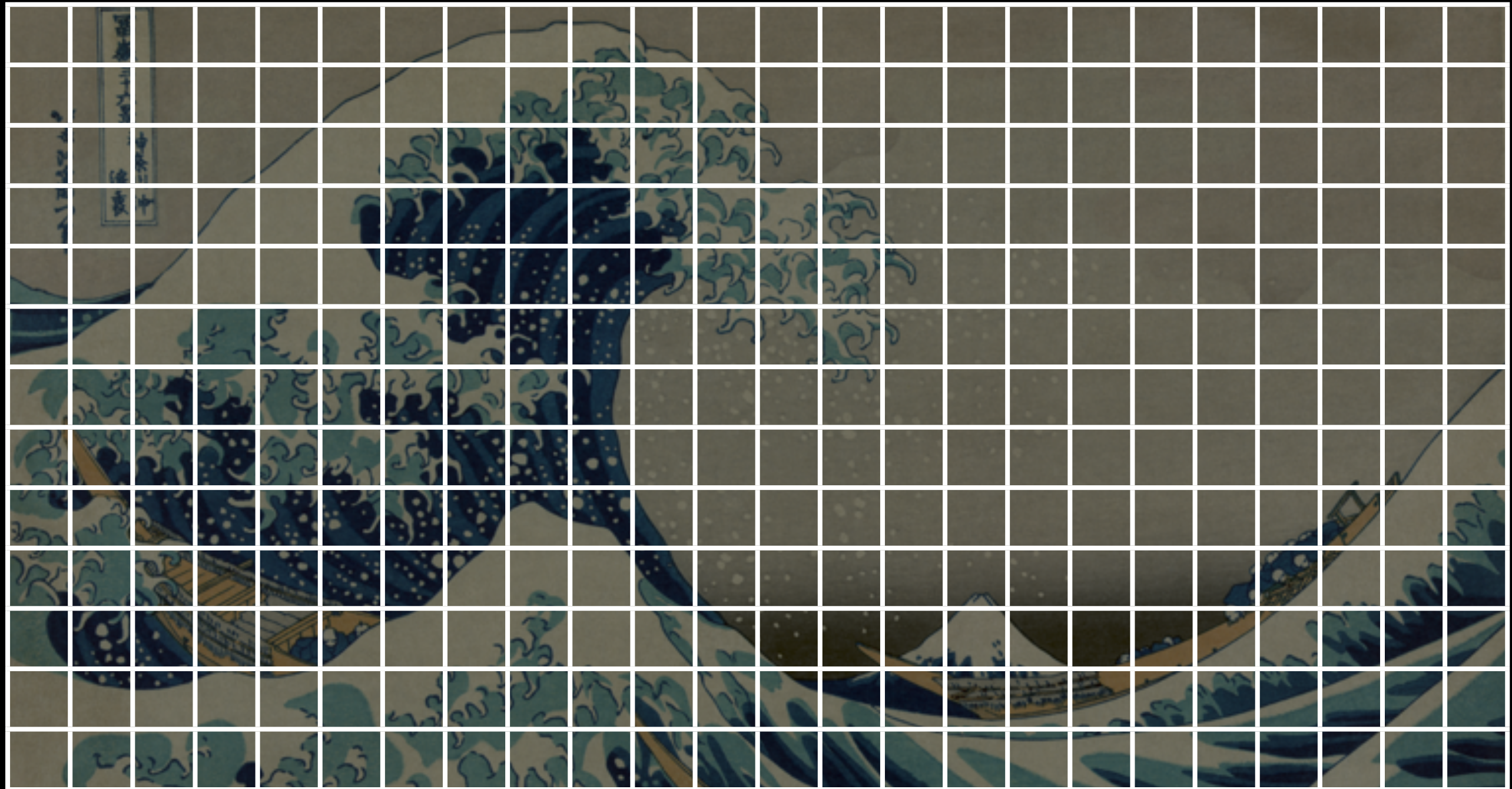
# Highly Adaptive Liquid Simulations on Tetrahedral Meshes

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ScanlineVFX

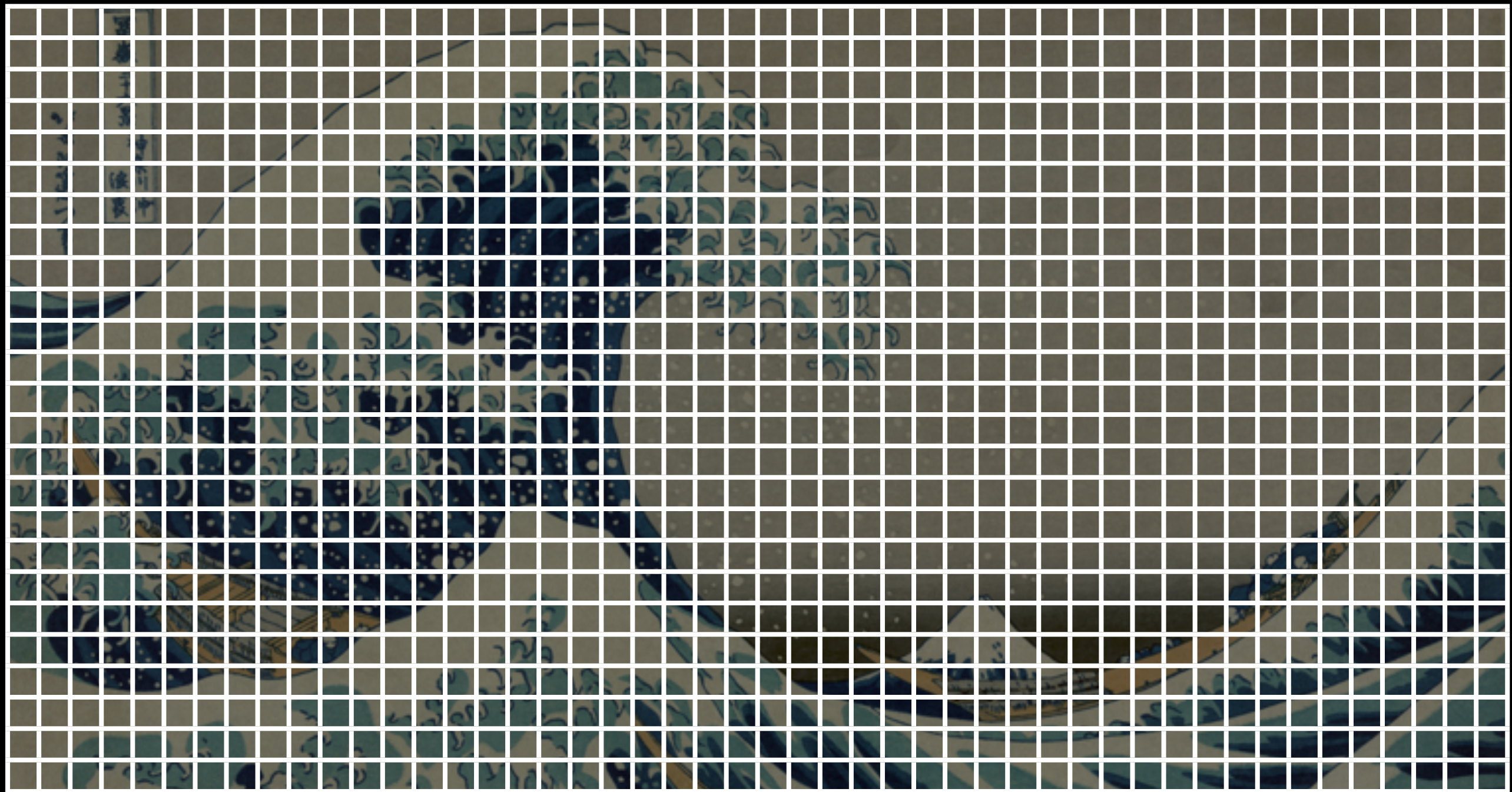
Chris Wojtan  
IST Austria

# Non-Adaptive Fluid Simulation



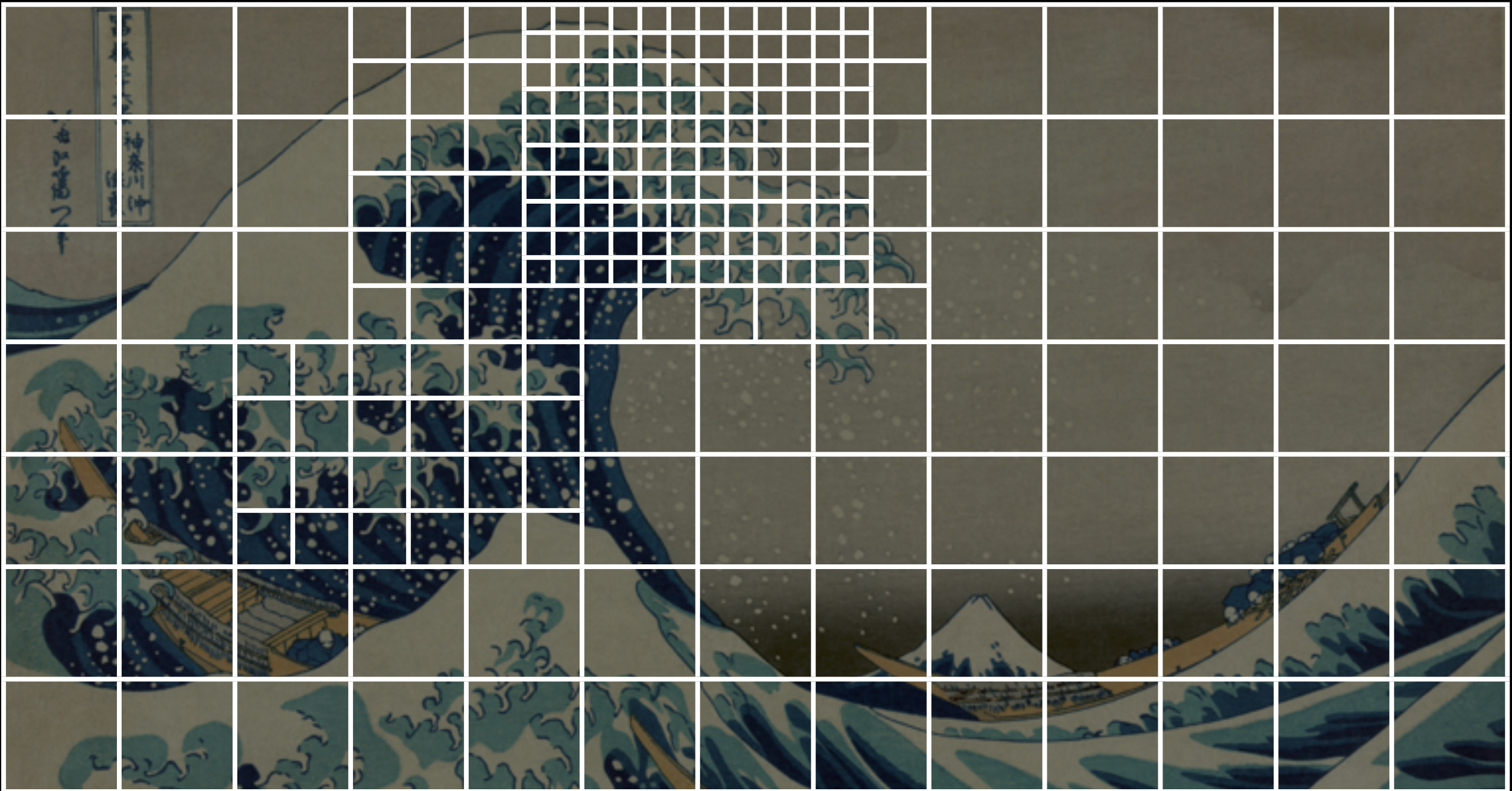


# Non-Adaptive Fluid Simulation



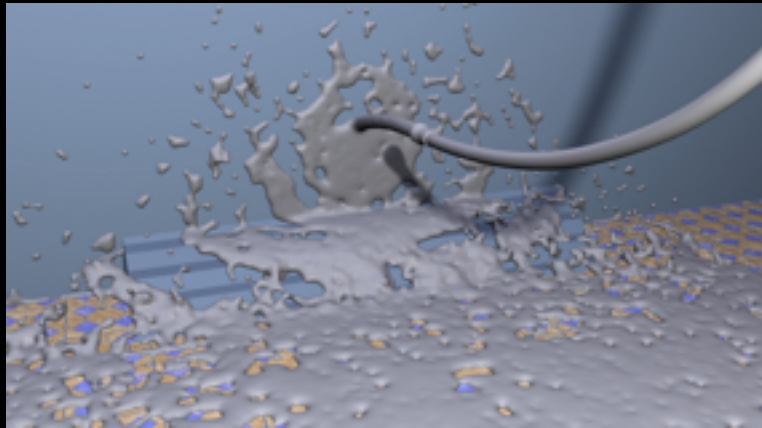


# Adaptive Fluid Simulation





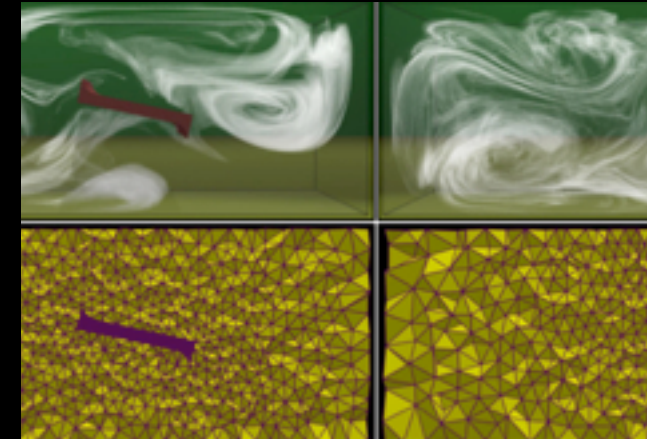
# Related work



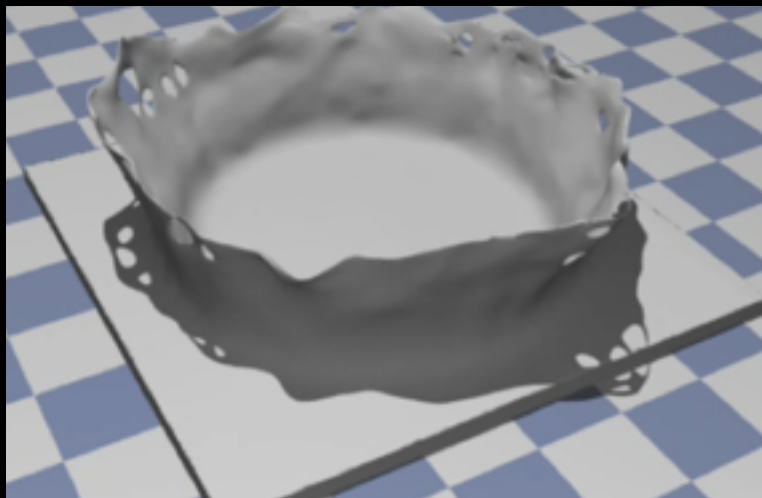
[Sin et al. 09]



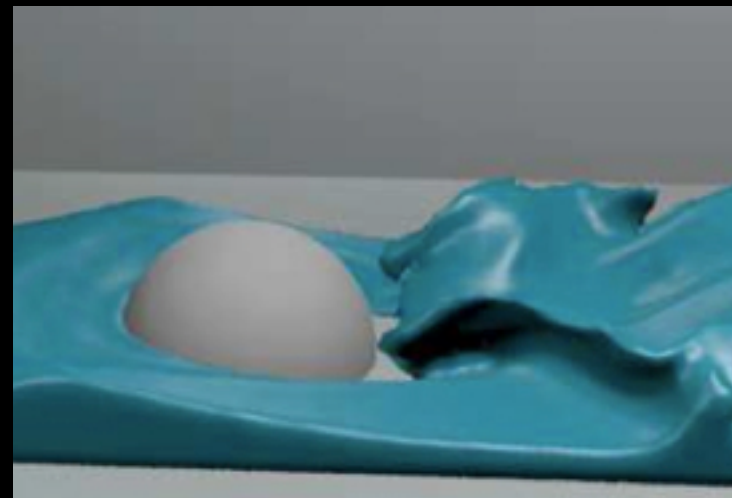
[Feldman et al. 05]



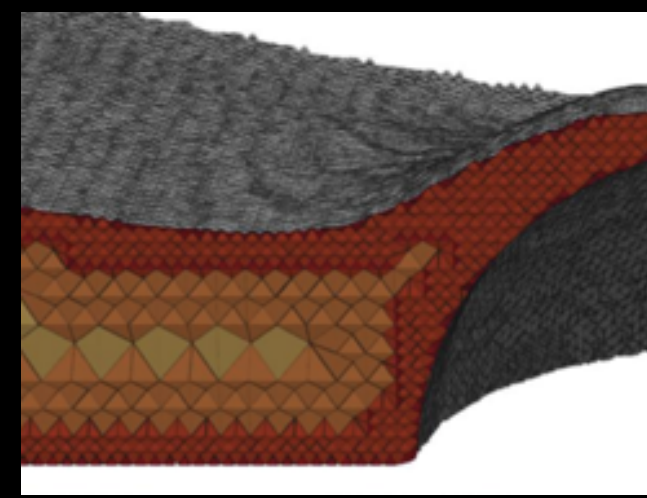
[Klingner et al. 06]



[Brochu et al. 10]



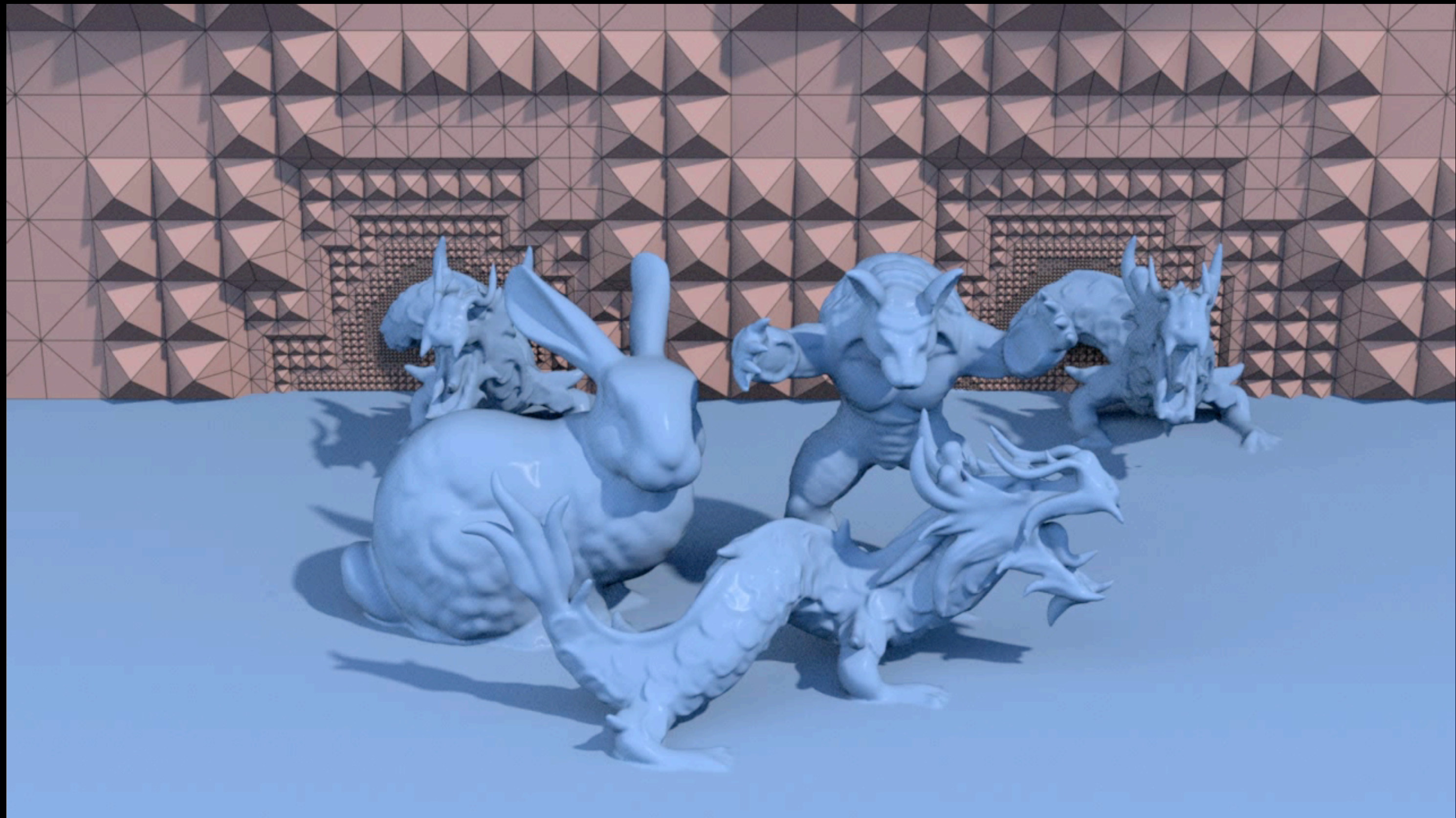
[Batty et al. 10]



[Chentanez et al. 07]







# Contribution

- Compact pressure discretization
- 2nd order free surface boundary condition
- Surface reconstruction for particles
- Highly adaptive sizing function



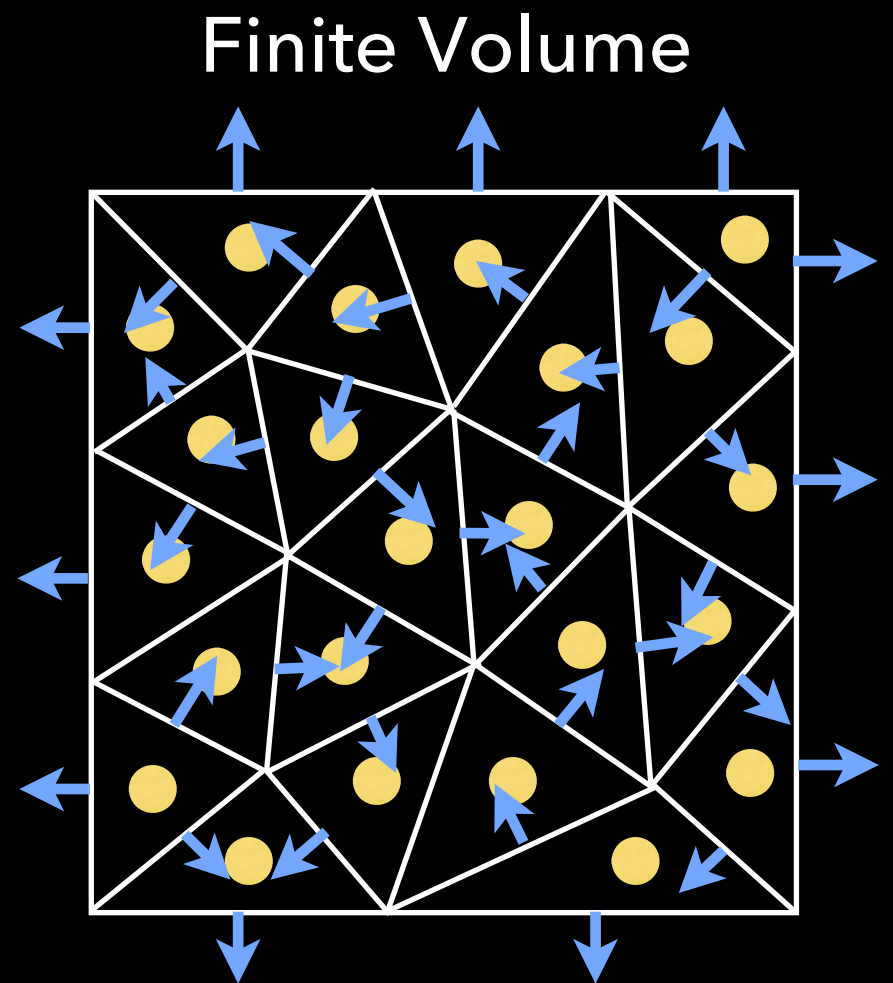
# Discretization

# Problem Statement

$$\frac{\Delta t}{\rho} \nabla^2 p - \nabla \cdot \boldsymbol{u}^* = 0$$



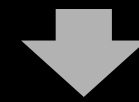
# Finite Volume Approach



Velocity: Facet 

Pressure: Center 

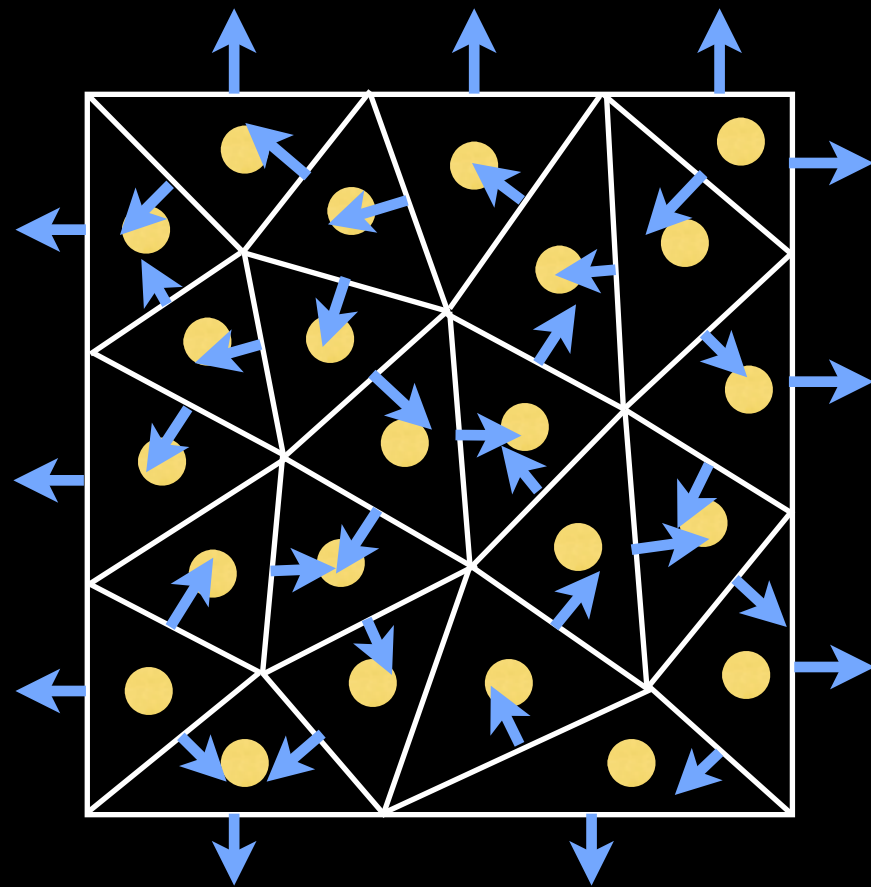
$$\frac{\Delta t}{\rho} \nabla^2 p - \nabla \cdot \mathbf{u}^* = 0$$



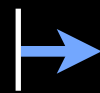
$$\oint_{d\Omega} \left( \frac{\Delta t}{\rho} \nabla p - \mathbf{u}^* \right) \cdot \mathbf{n} = 0$$

# Finite Volume & Our Approach

Finite Volume



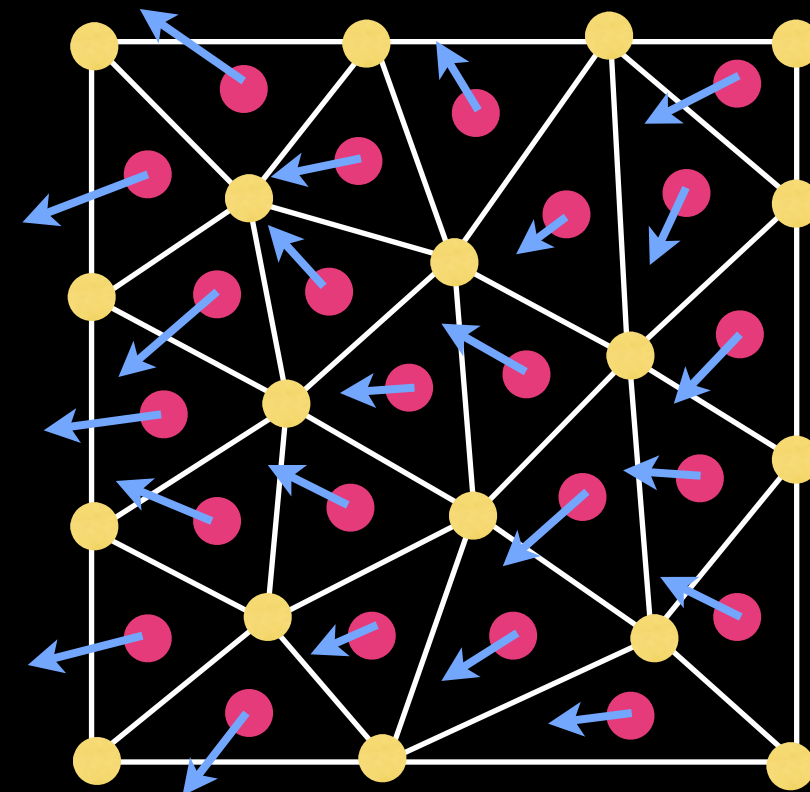
Velocity: Facet



Pressure: Center



Our Method



Velocity: Center



Pressure: Node





# Our Approach

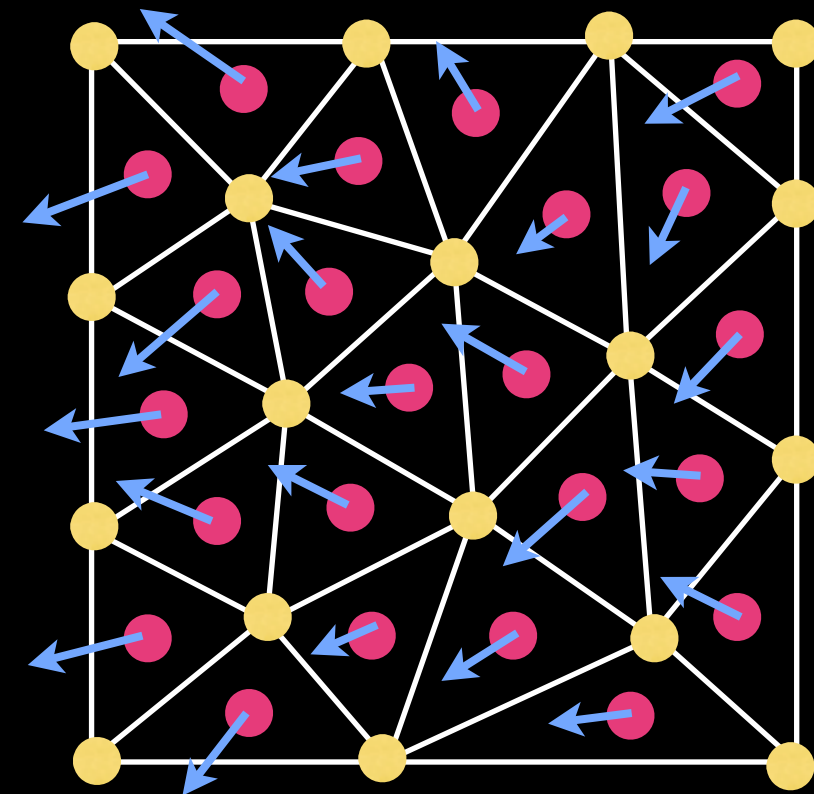
$$\frac{\Delta t}{\rho} \nabla^2 p - \nabla \cdot \mathbf{u}^* = 0$$



$$p = \arg \min_p \int_{\Omega} \frac{1}{2} \left\| \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p \right\|^2 \rho dV$$

[Batty et al. 07]

## Our Method



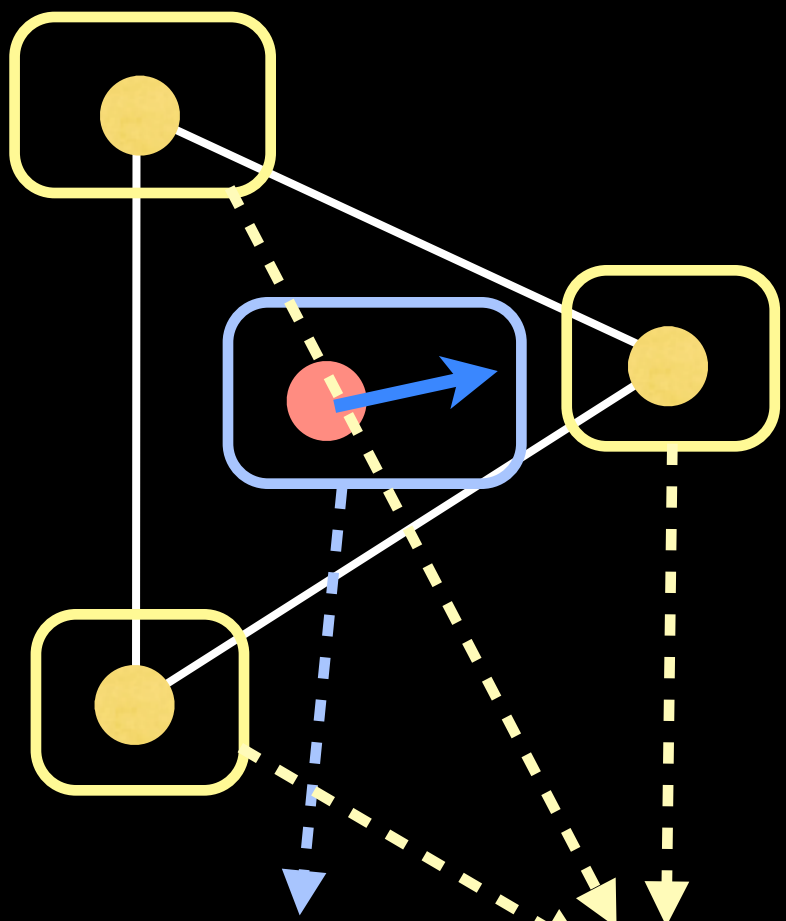
Velocity: Center



Pressure: Node



# Our Approach



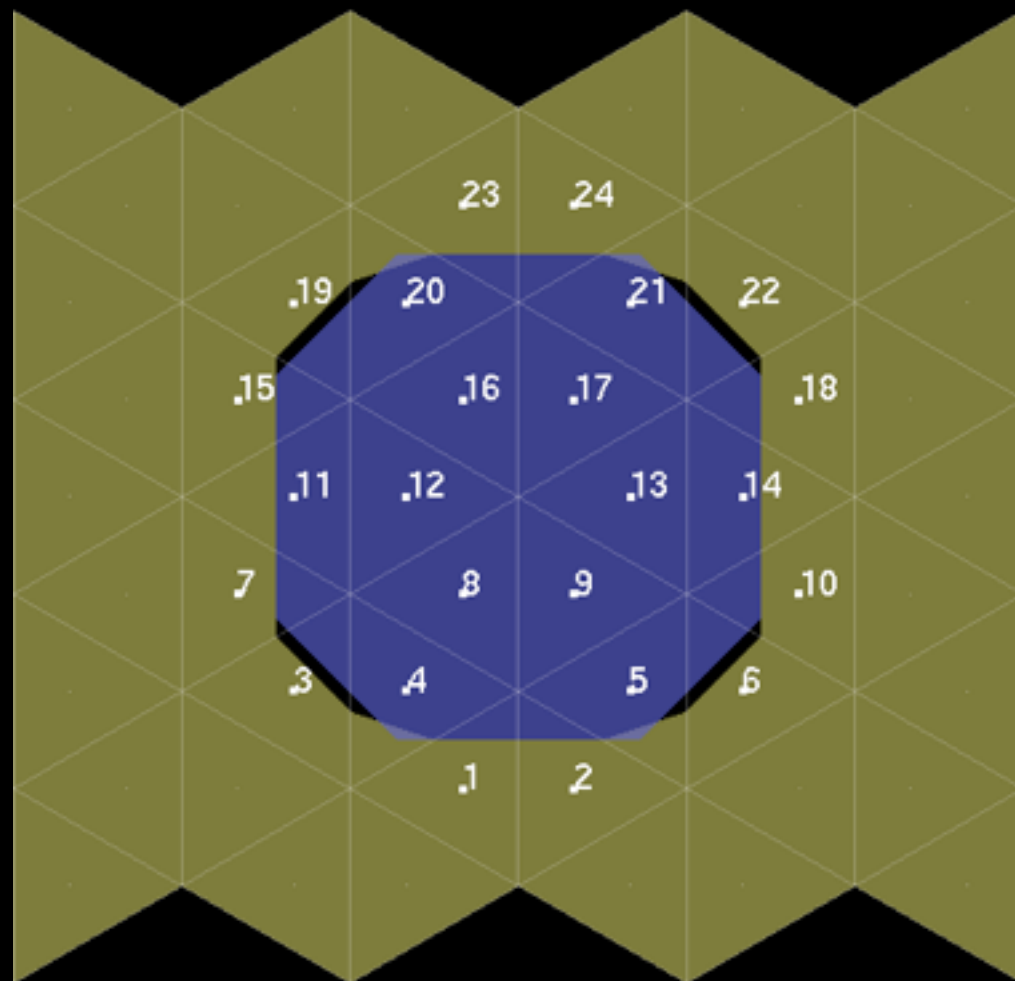
The diagram illustrates a network structure with four yellow nodes (circles in boxes) and one central red node (circle in a blue box). The red node has a blue arrow pointing right. Dashed lines connect the nodes to a mathematical expression at the bottom. The expression is:

$$p = \arg \min_p \int_{\Omega} \frac{1}{2} \left\| \boxed{u^*} - \frac{\Delta t}{\rho} \boxed{\nabla p} \right\|^2 \rho dV$$

The terms  $\boxed{u^*}$  and  $\boxed{\nabla p}$  are highlighted with blue and yellow boxes, respectively, corresponding to the colors in the diagram above.

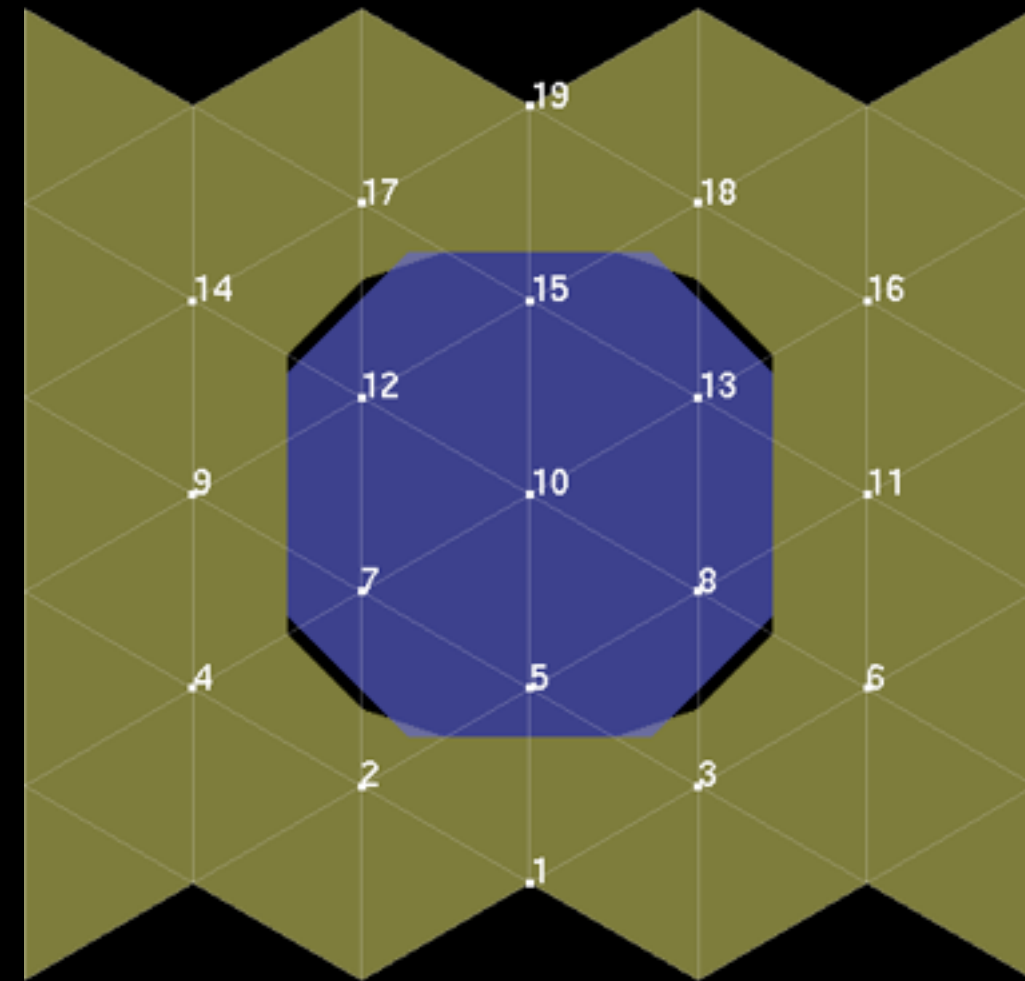
# Fewer Pressure Samples

Finite Volume



24 Pressure values

Our Method

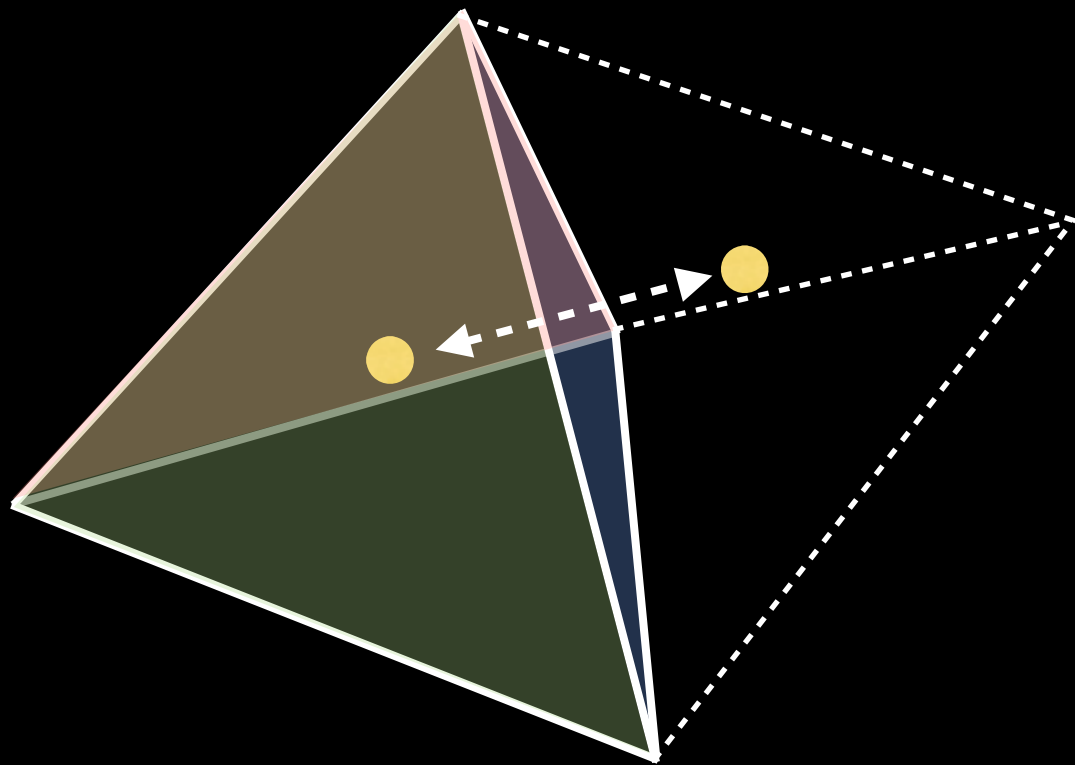


19 Pressure values

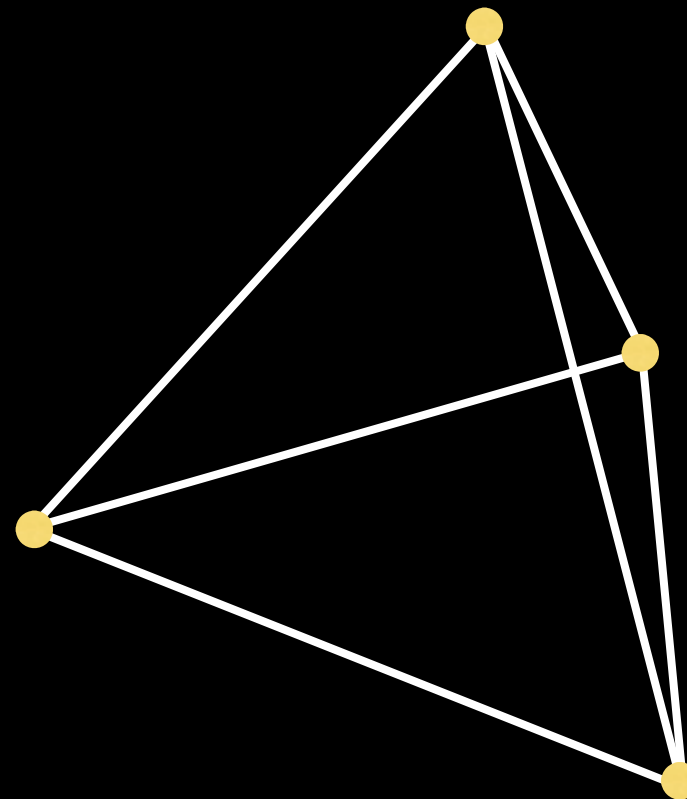


# Simpler Bookkeeping

Finite Volume

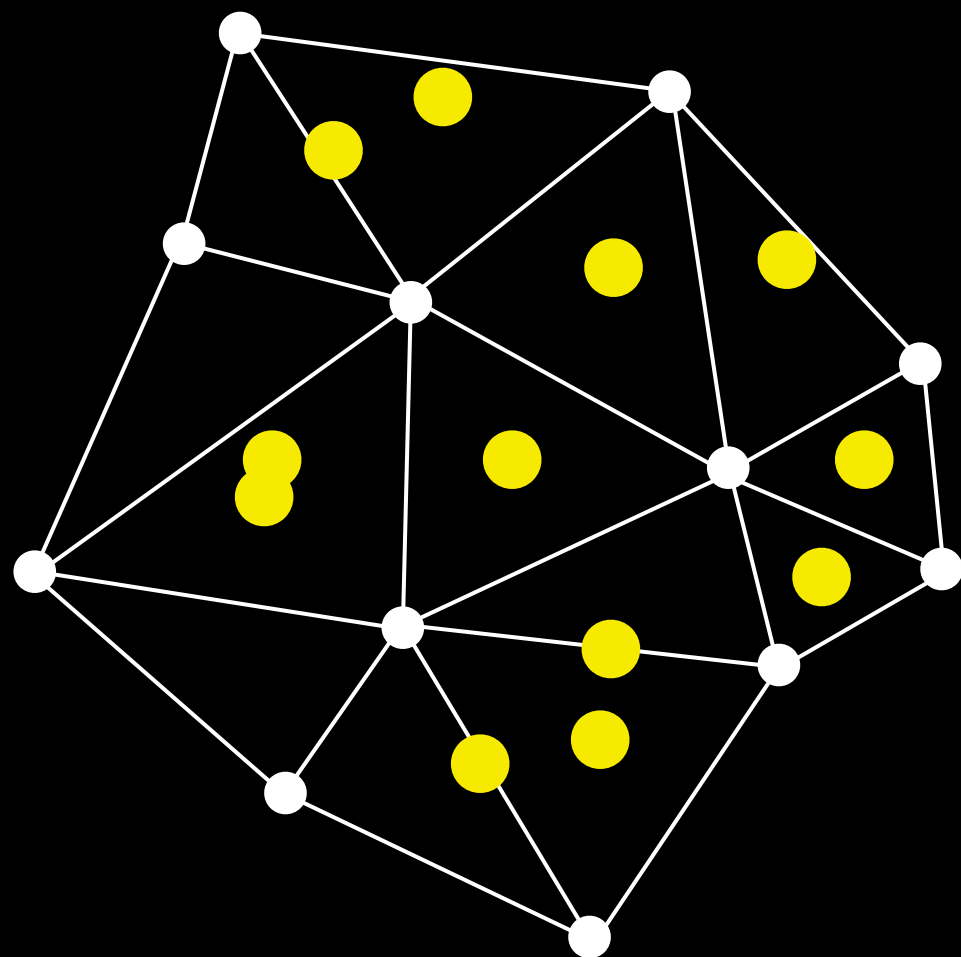


Our Method

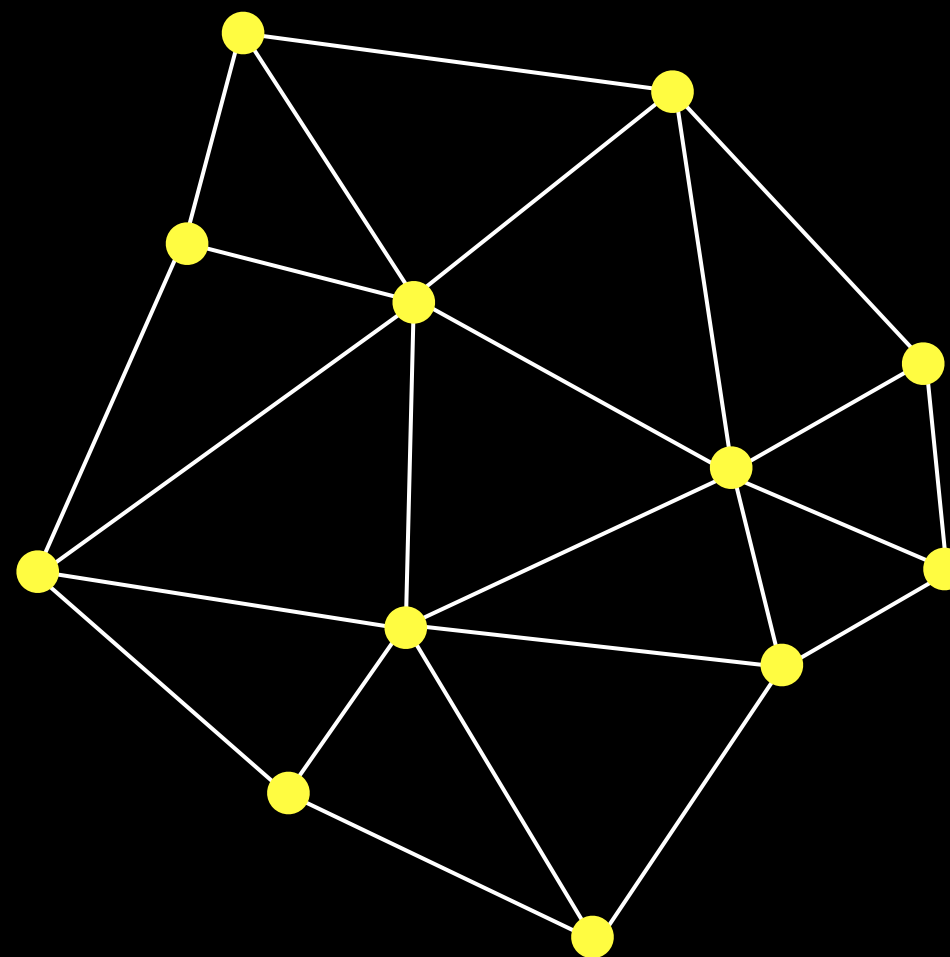


# Numerical Robustness

Finite Volume



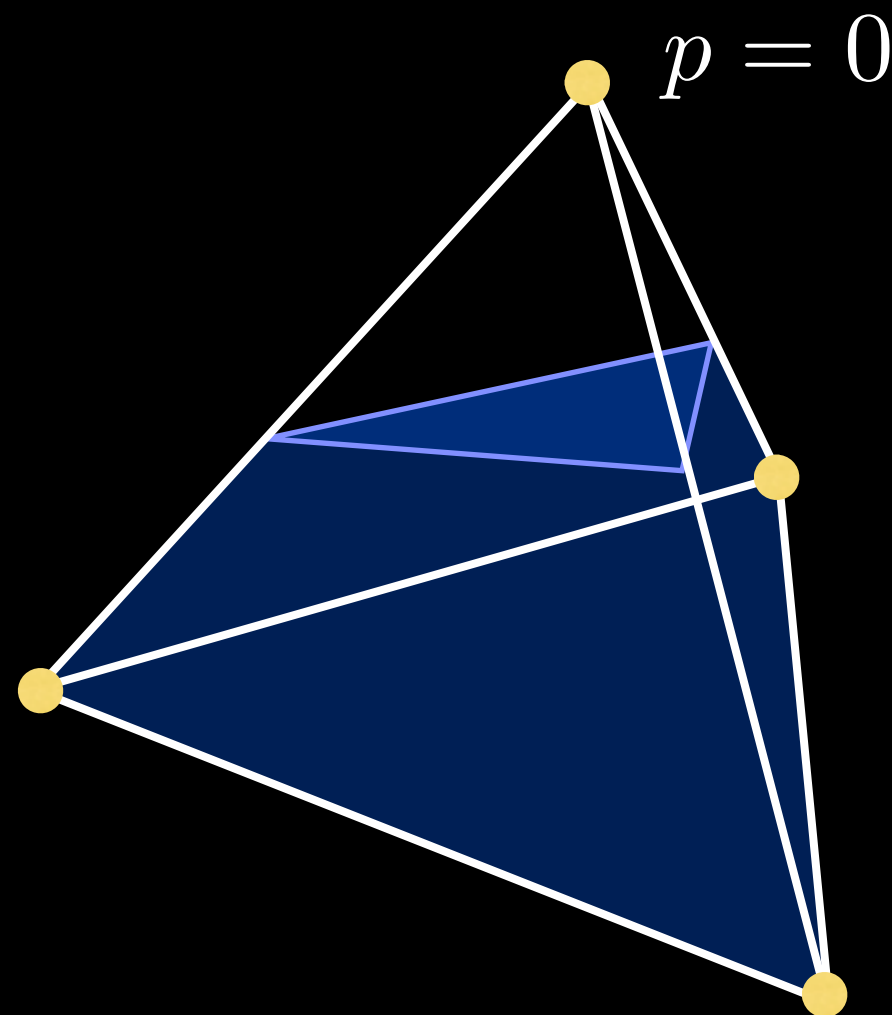
Our Method



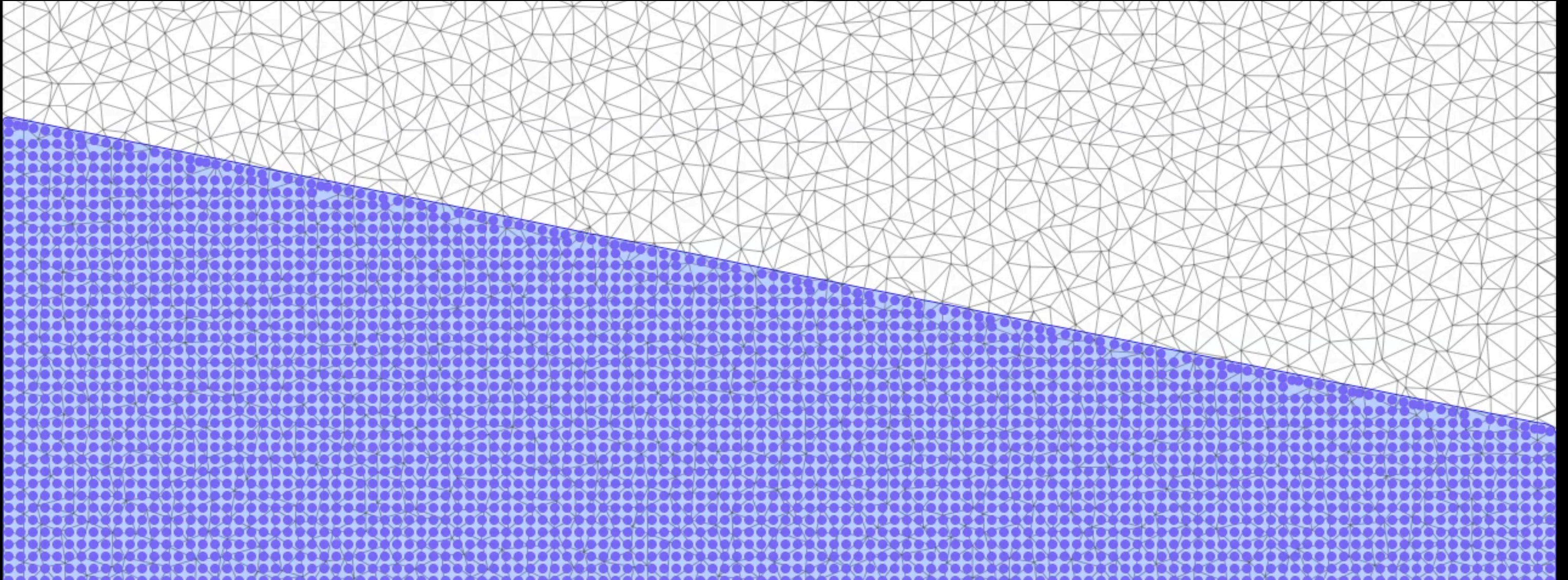
Boundary Condition



1st order accurate

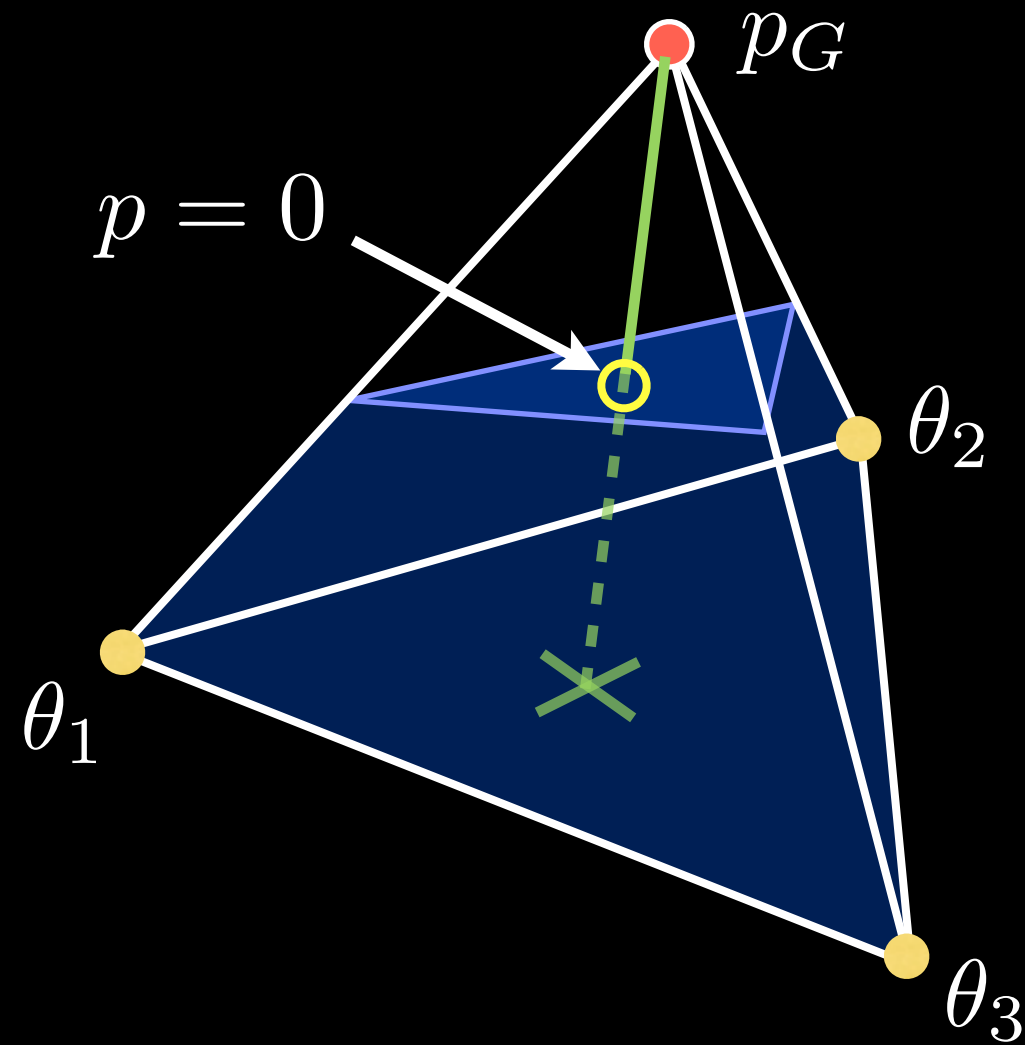


1st order accurate

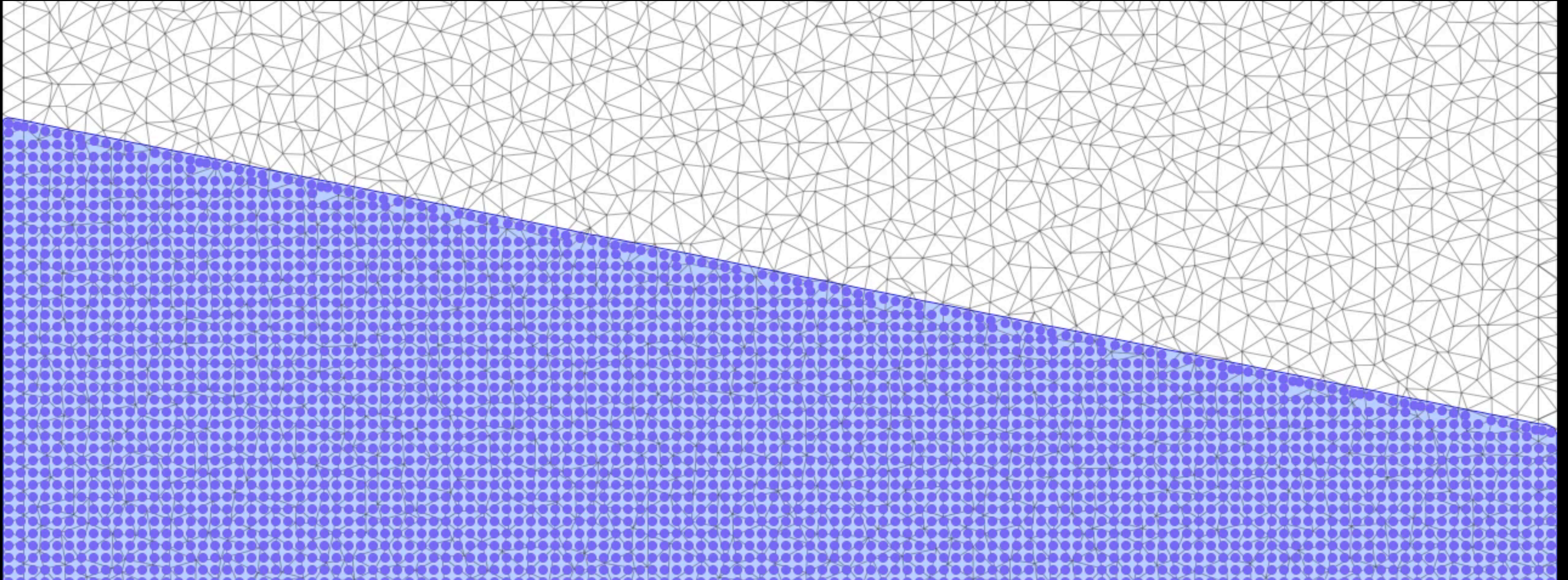




2nd order accurate



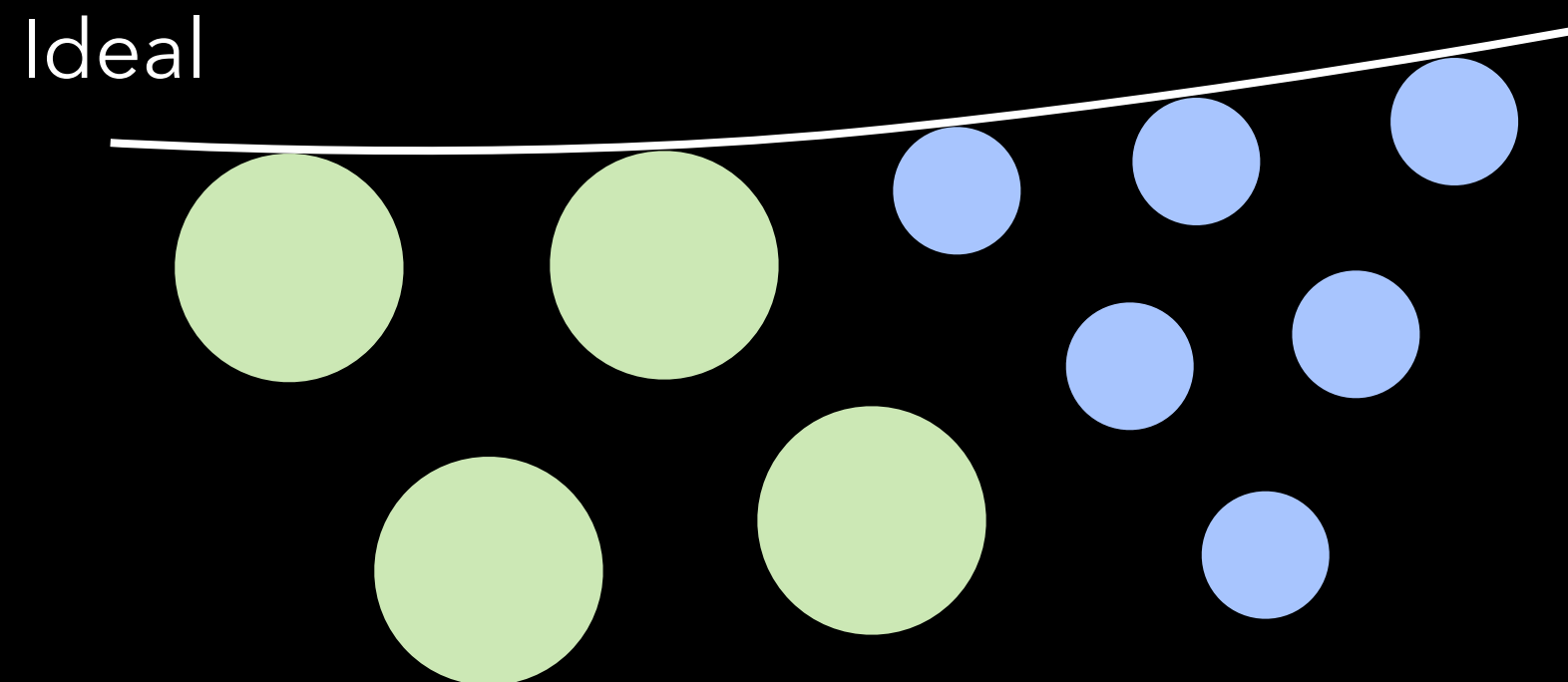
2nd order accurate





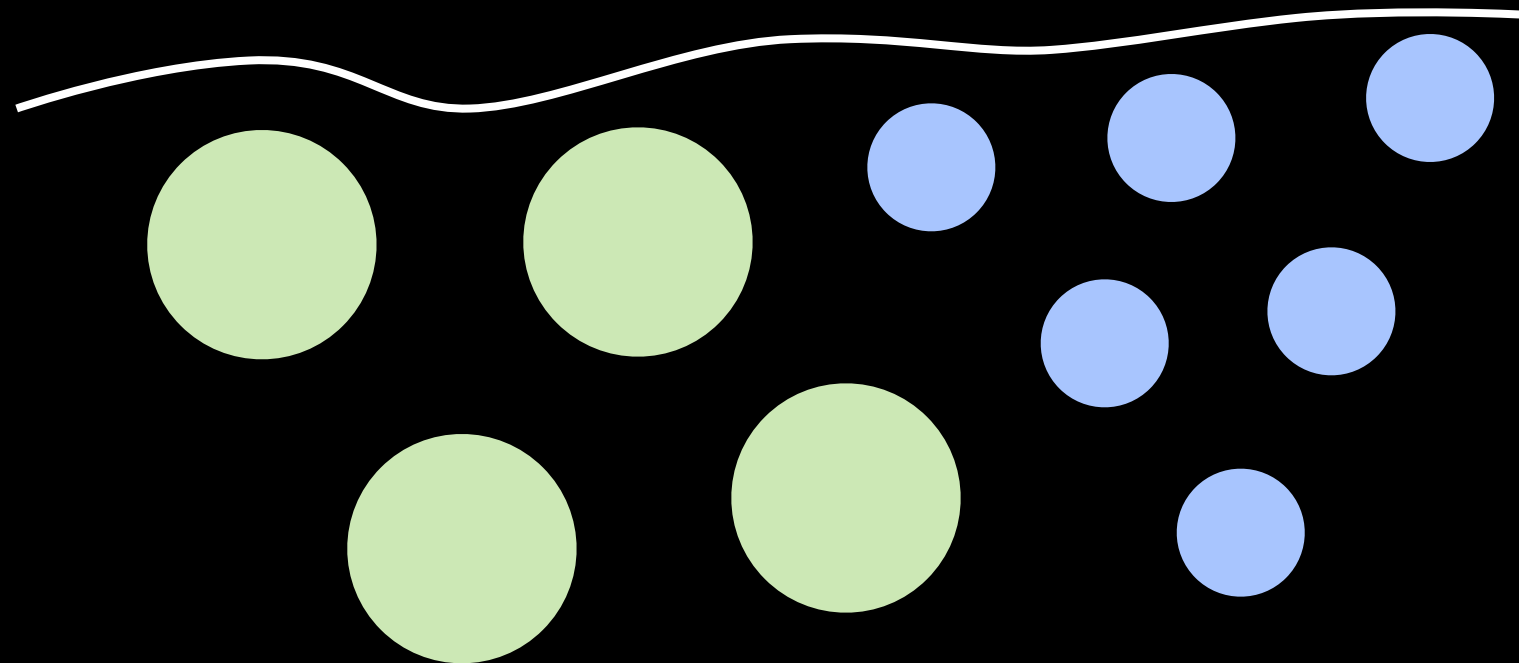
# Surface Reconstruction

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# Surface Reconstruction

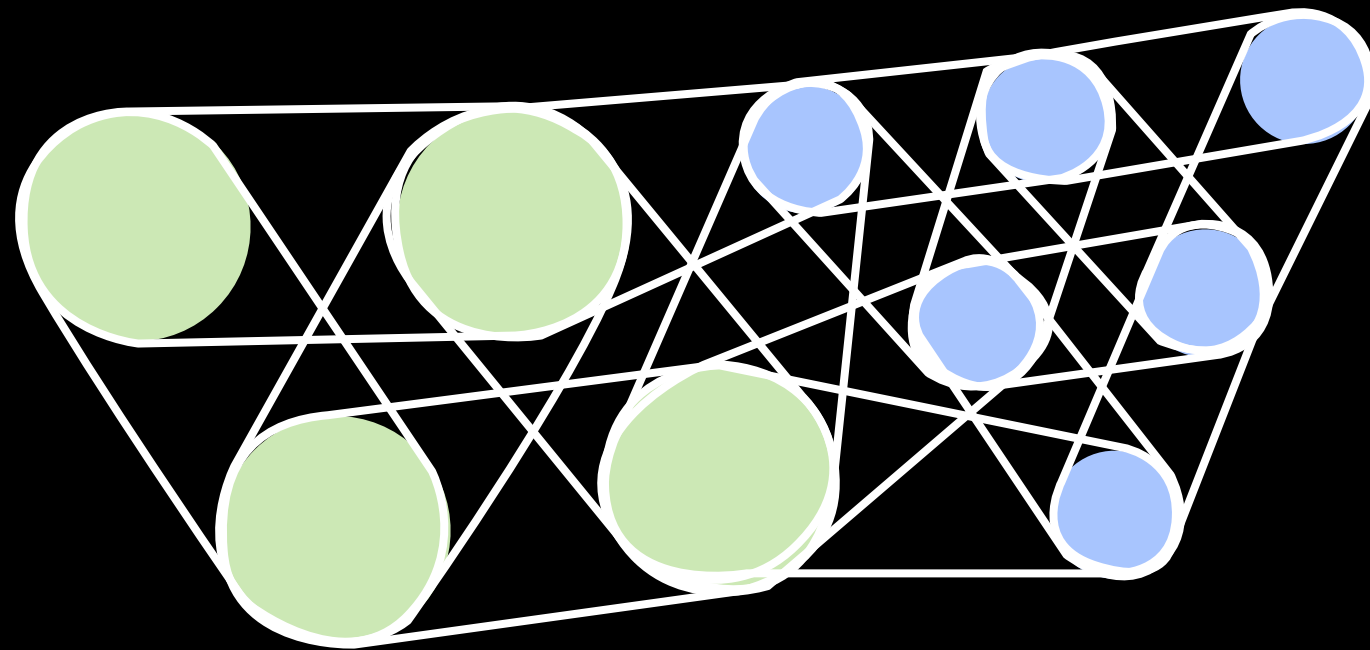
Previous Approaches





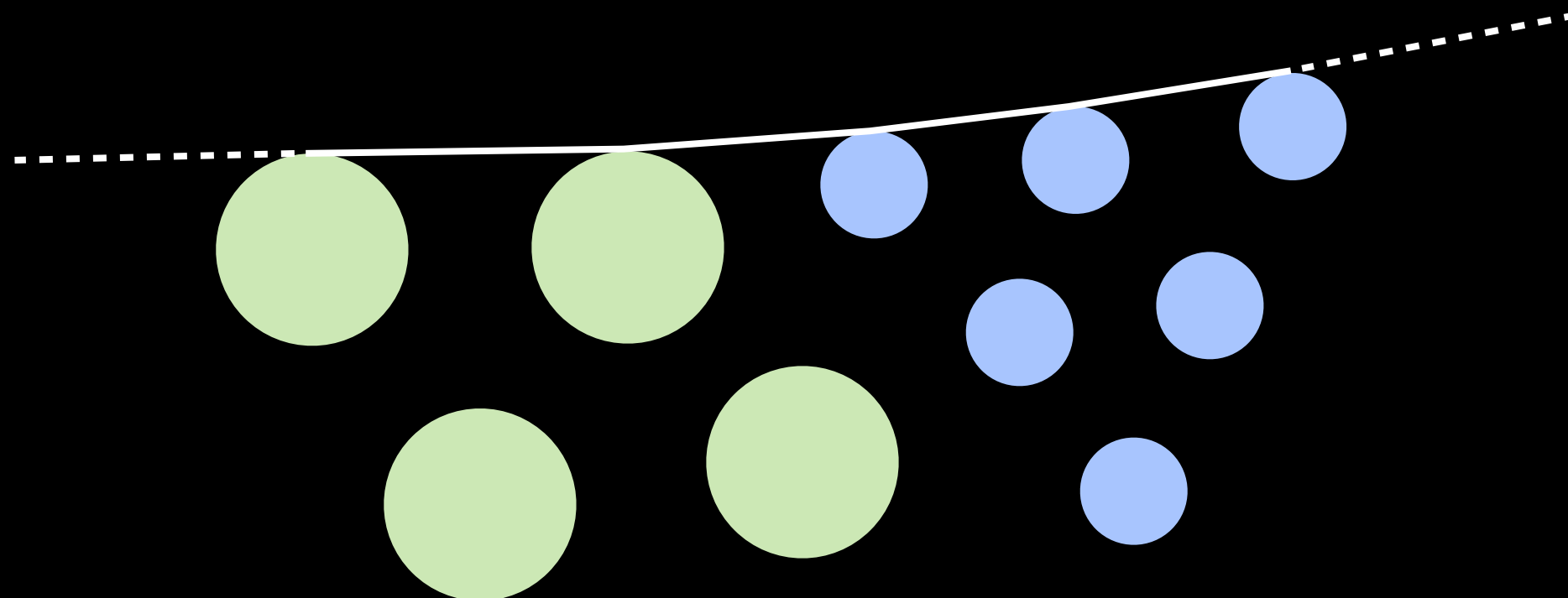
# Surface Reconstruction

Our Approach

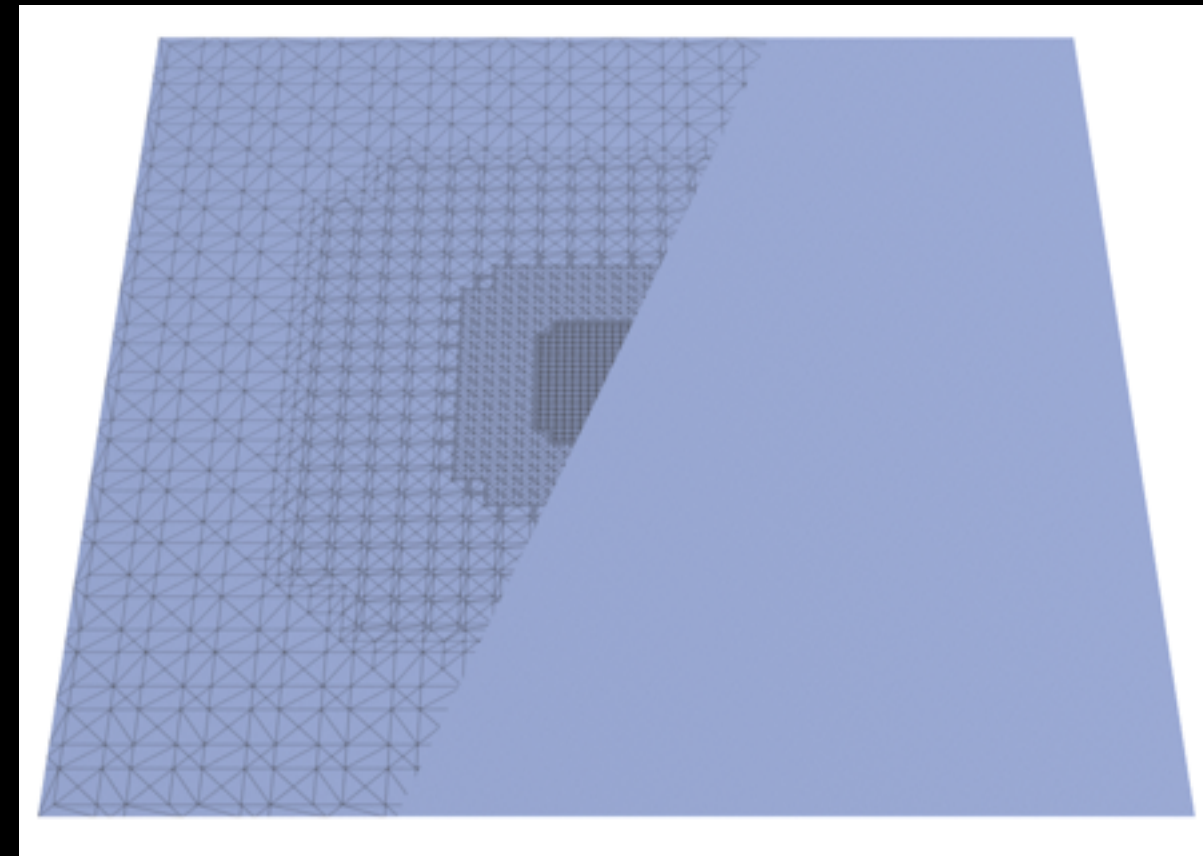
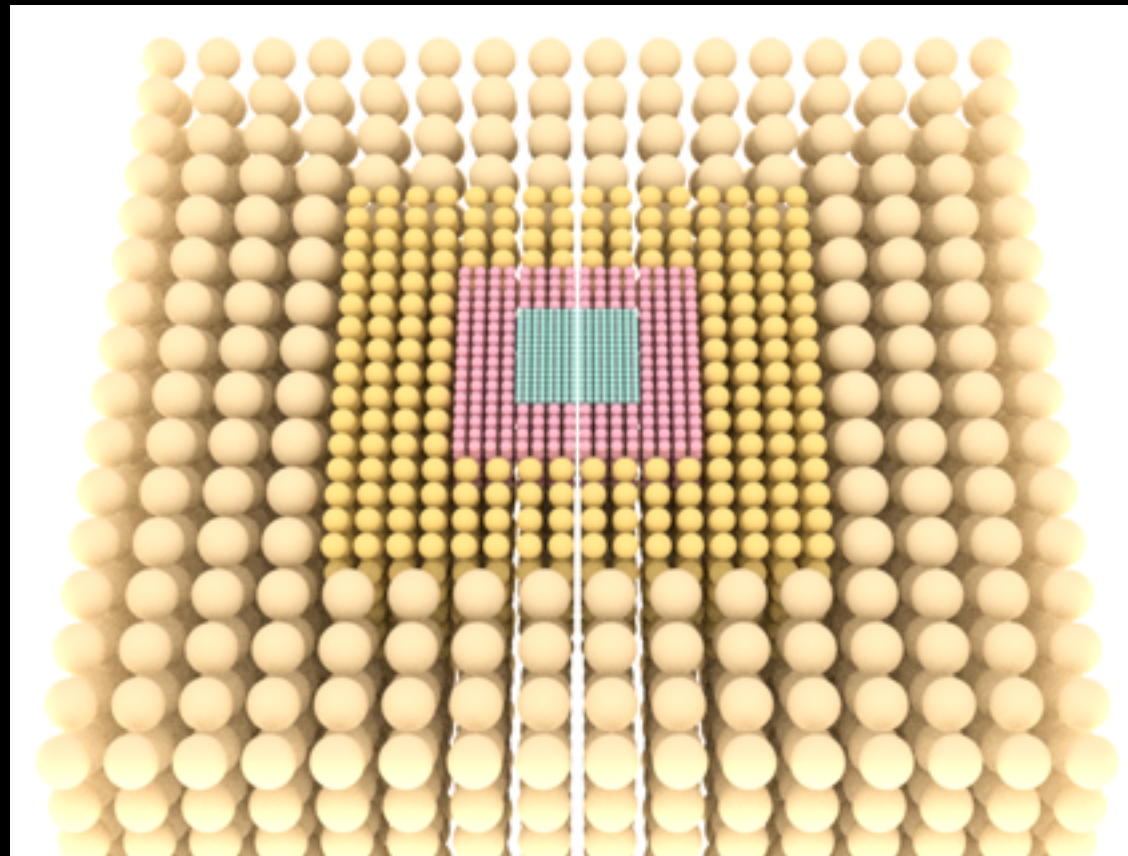


# Surface Reconstruction

Our Approach



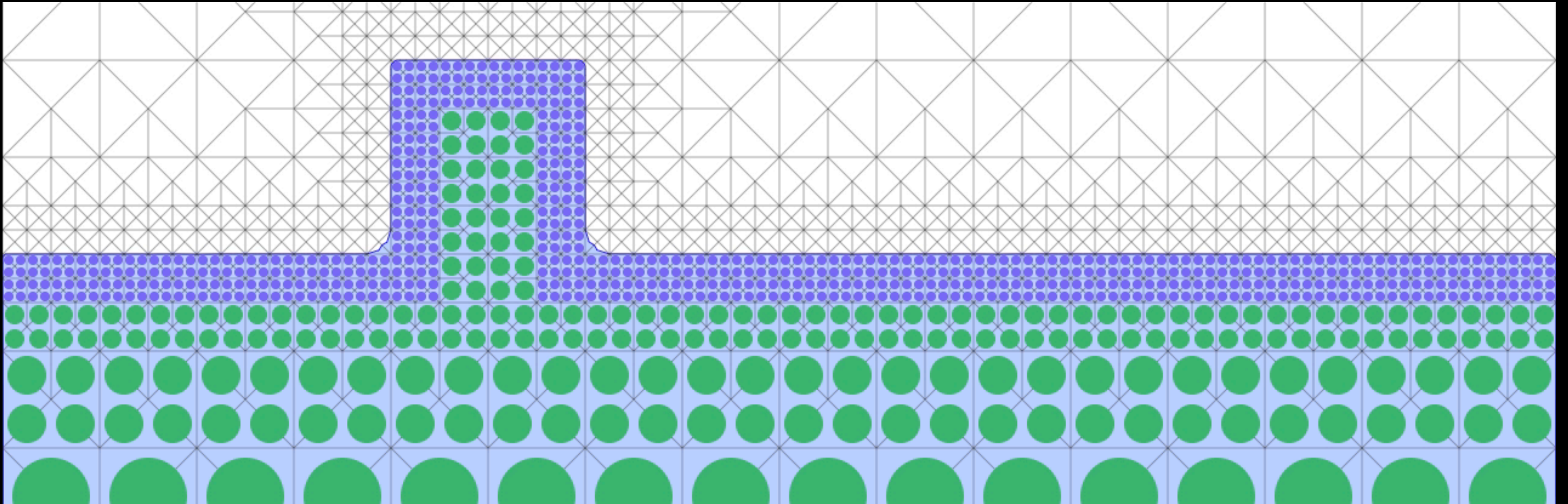
# Surface Reconstruction



Sizing function

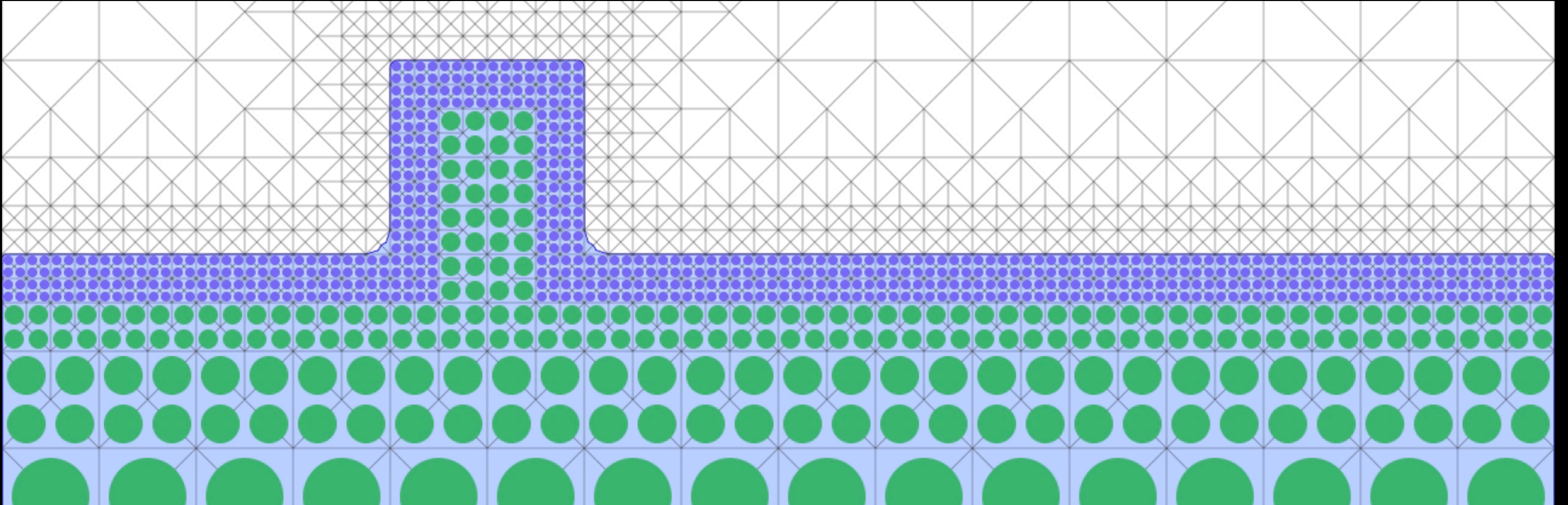


# Sizing function



Depth

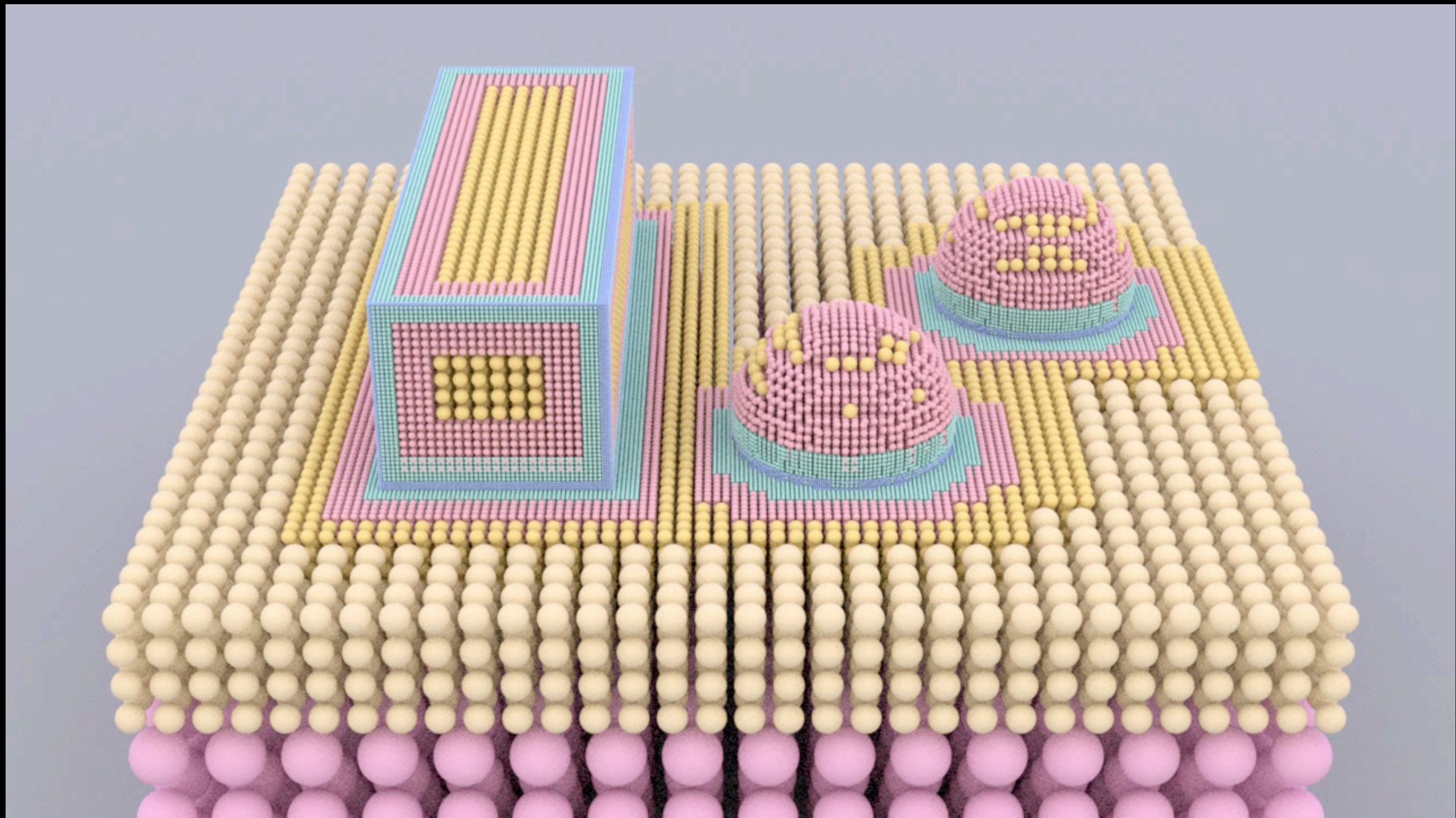
# Sizing function



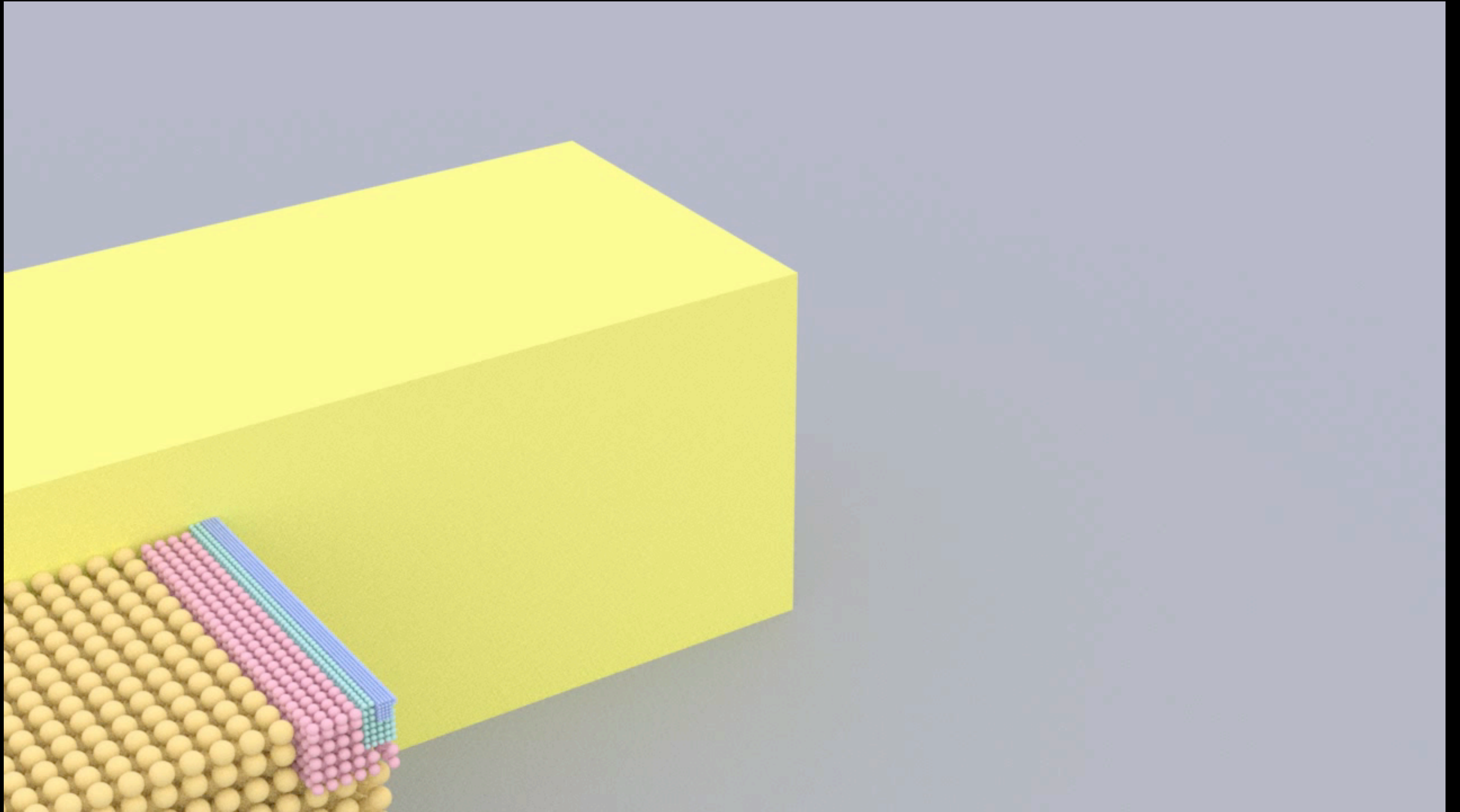
Curvature, Strain, Depth

# Results

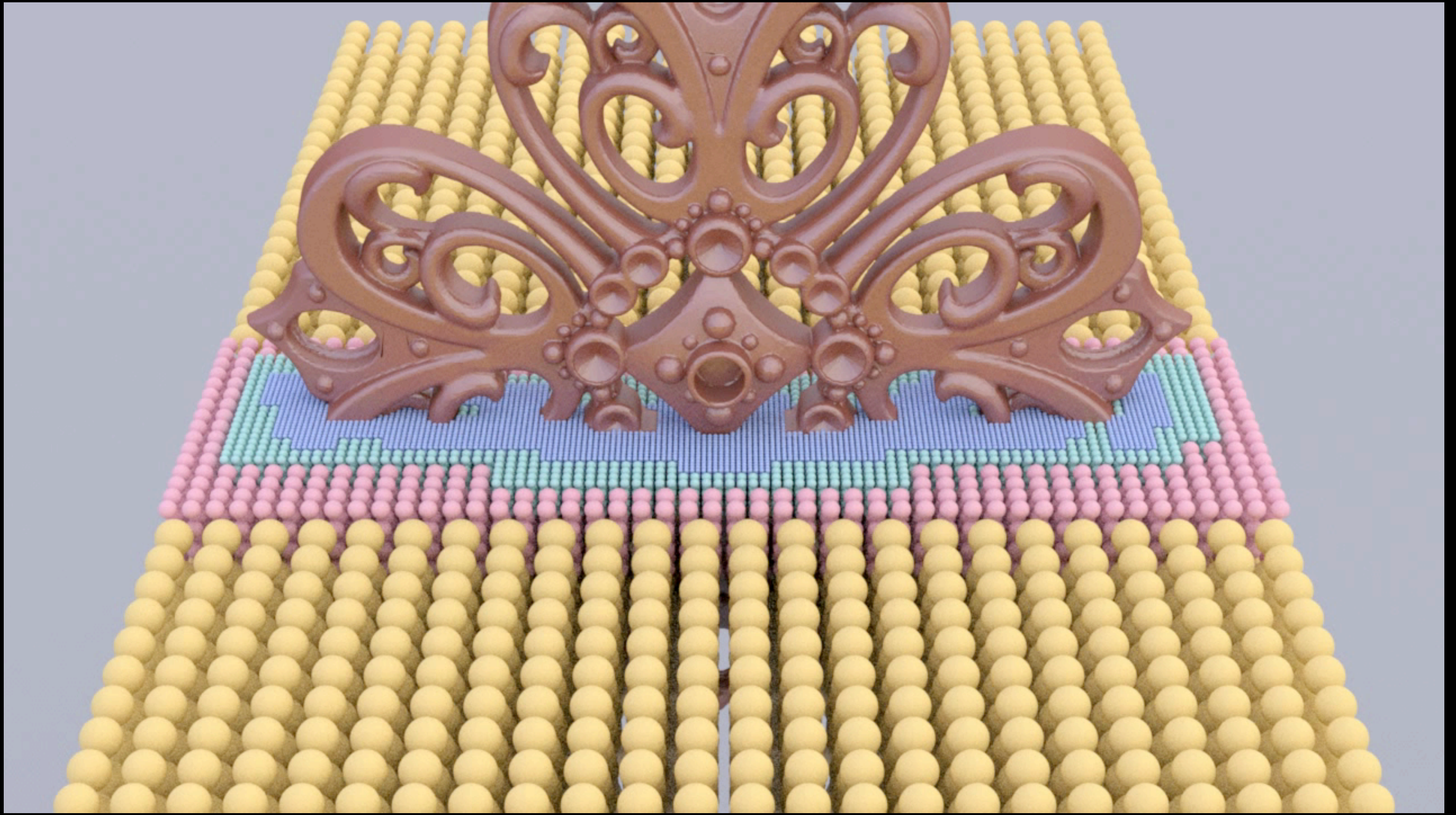




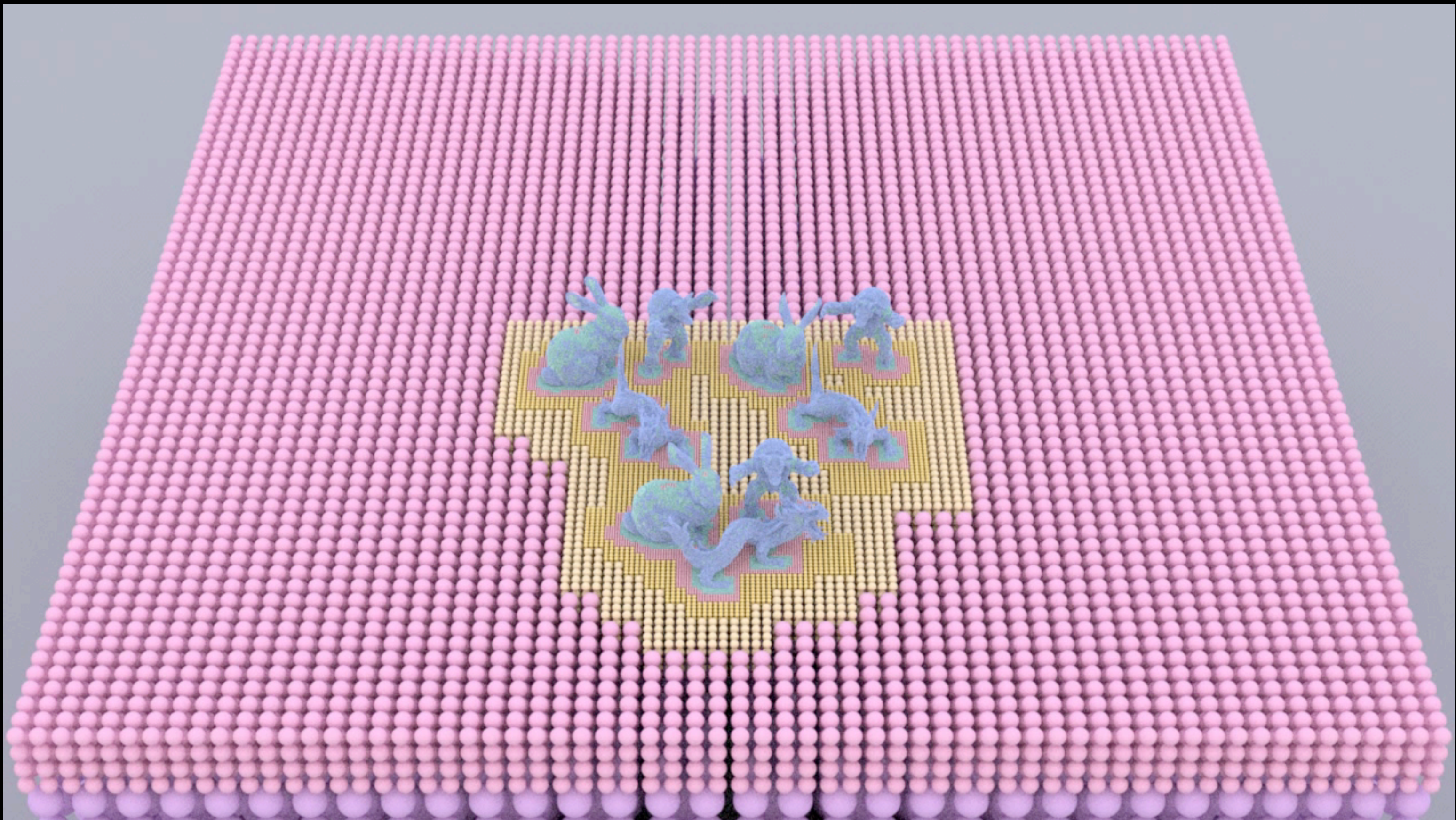












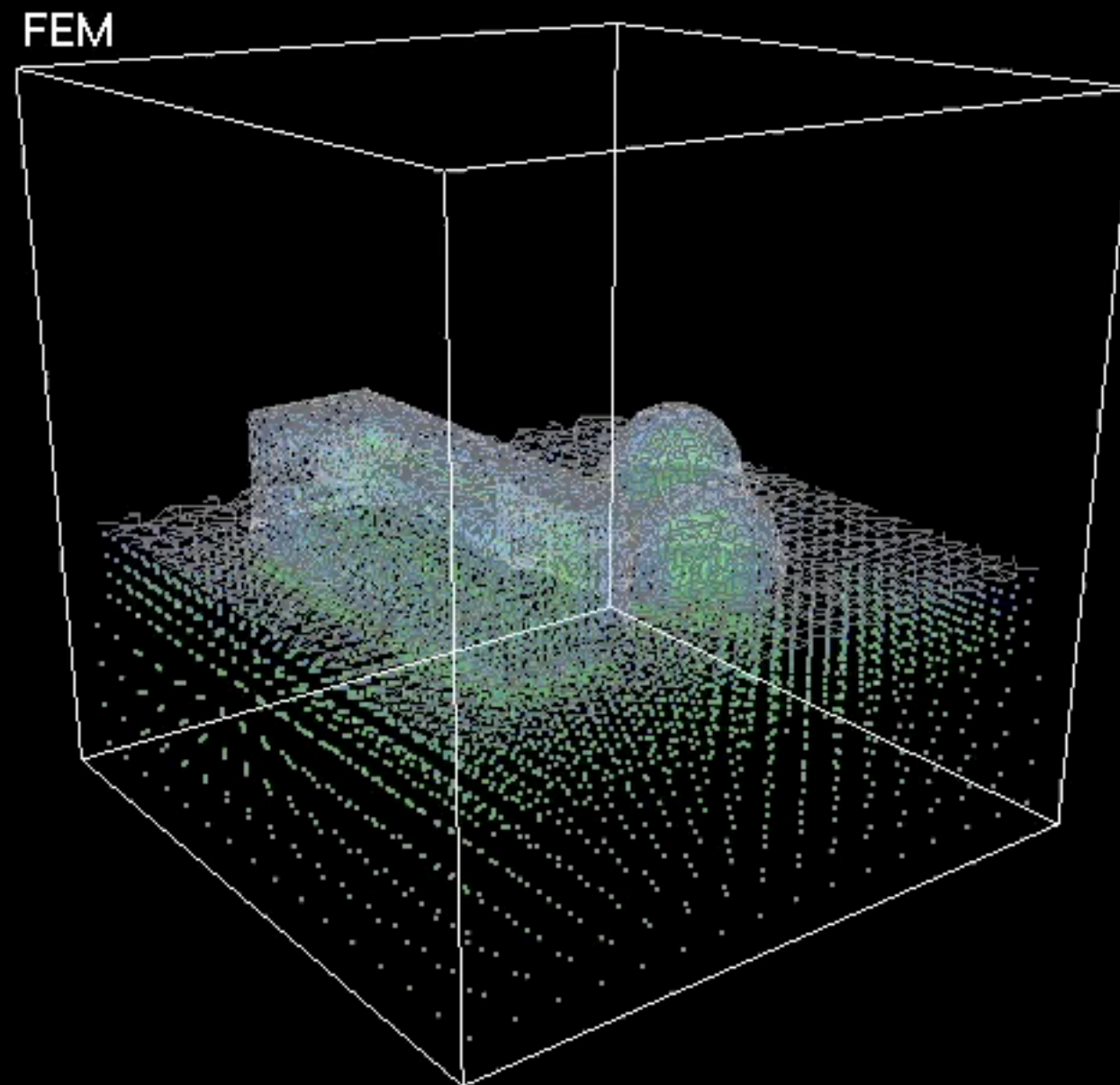


# Limitations

- Slow surface reconstruction
- Artifacts when graded too quickly
- Pressure overlooks high frequency velocity



# Code



[http://pub.ist.ac.at/group\\_wojtan/projects/2013\\_Ando\\_HALSoTM/](http://pub.ist.ac.at/group_wojtan/projects/2013_Ando_HALSoTM/)

# Summary

- Tetrahedral pressure solver
- Accurate boundary condition
- Surface reconstruction for particles at different sizes
- Flexible sizing function

Thank you !